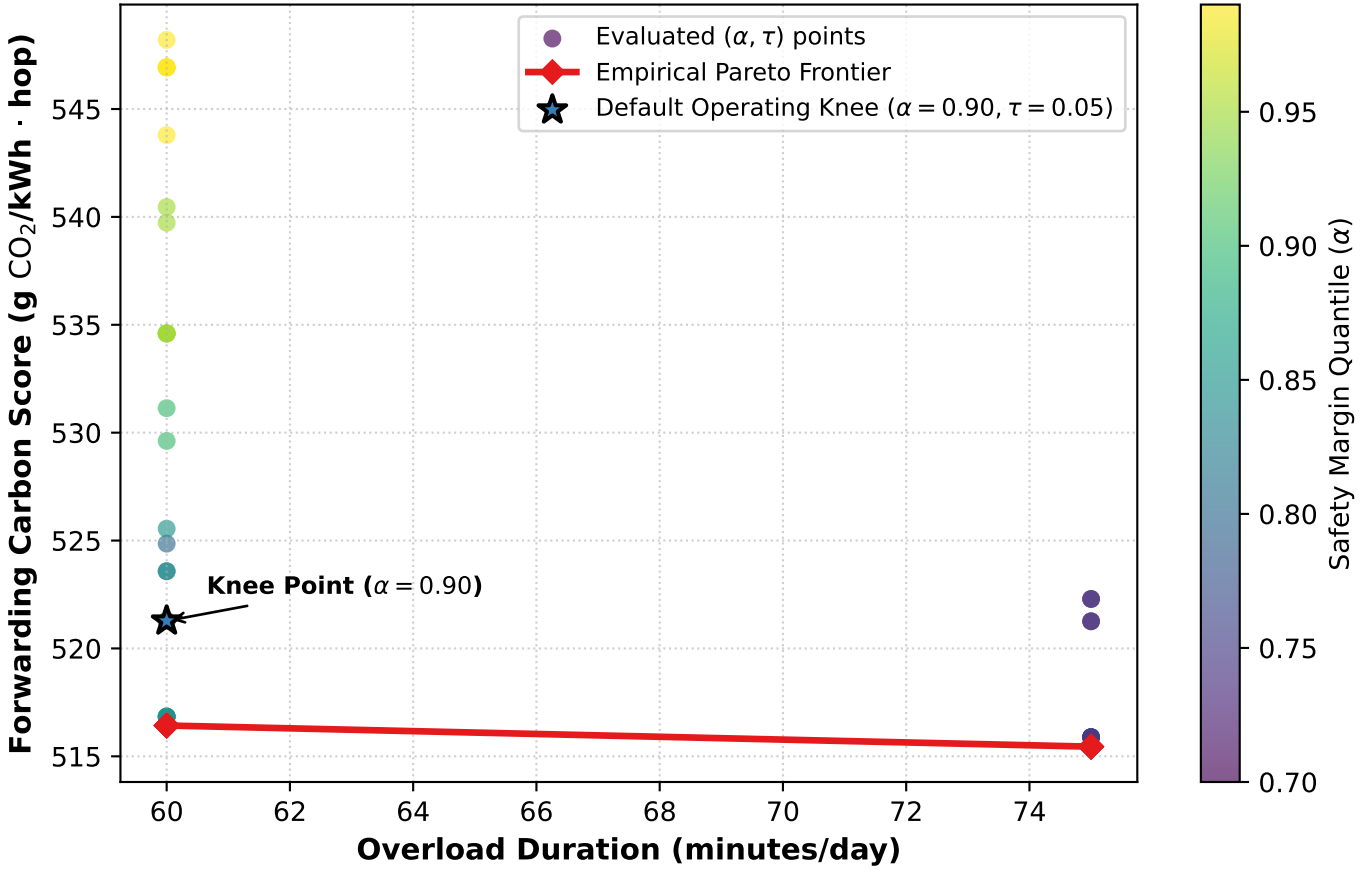
Empirical Pareto Trade-off: Carbon vs. Overload Duration
Evaluated (α, τ) points
Empirical Pareto Frontier
Default Operating Knee (α = 0.90, τ = 0.05)
Knee Point (α = 0.90)
Forwarding Carbon Score (g CO2/kWh · hop)
Overload Duration (minutes/day)
Safety Margin Quantile (α)
515
520
525
530
535
540
545
60
62
64
66
68
70
72
74
0.70
0.75
0.80
0.85
0.90
0.95